

Question 1

Let $P(n)$ be the statement that $1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$ for the positive integer n .

a) What is the statement $P(1)$? $P(1)$ is the statement that $1^3 = \left(\frac{1(1+1)}{2}\right)^2$.

b) Show that $P(1)$ is true, completing the basis step of the proof. $1^3 = 1 = \left(\frac{1(1+1)}{2}\right)^2 = \left(\frac{2}{2}\right)^2 = 1^2 = 1$.

c) What is the inductive hypothesis? The inductive hypothesis is that $1^3 + 2^3 + \dots + k^3 = \left(\frac{k(k+1)}{2}\right)^2$ for some positive integer k .

d) What do you need to prove in the inductive step? In the inductive step, we need to prove that $1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2}\right)^2$.

e) Complete the inductive step, identifying where you use the inductive hypothesis. Assume the inductive hypothesis: $1^3 + 2^3 + \dots + k^3 = \left(\frac{k(k+1)}{2}\right)^2$.

Then we need to show:

$$\begin{aligned} 1^3 + 2^3 + \dots + k^3 + (k+1)^3 &= \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 \\ &= \frac{k^2(k+1)^2}{4} + (k+1)^3 \\ &= \frac{k^2(k+1)^2 + 4(k+1)^3}{4} \\ &= \frac{(k+1)^2(k^2 + 4(k+1))}{4} \\ &= \frac{(k+1)^2(k^2 + 4k + 4)}{4} \\ &= \frac{(k+1)^2(k+2)^2}{4} \\ &= \left(\frac{(k+1)(k+2)}{2}\right)^2. \end{aligned}$$

Thus, $1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2}\right)^2$.

This completes the inductive step and proves the statement by mathematical induction.

Question 2

Prove that $1^2 + 3^2 + 5^2 + \dots + (2n + 1)^2 = (n + 1)(2n + 1)\frac{2n+3}{3}$

Base Case: $P(0)$ is the statement that $(0n + 1)^2 = 1^2 = (0 + 1)(0 + 1)\frac{2\cdot 0+3}{3}$.

$$1^2 = 1 = (1)(1)\frac{2\cdot 0+3}{3} = 1 \cdot 1 \cdot 1 = 1 \implies P(0) \text{ is true.}$$

Inductive Hypothesis: The inductive hypothesis is that $1^2 + 3^2 + 5^2 + \dots + (2k + 1)^2 = (k + 1)(2k + 1)\frac{2k+3}{3}$ for some positive integer k .

In the inductive step, we need to prove that $1^2 + 3^2 + 5^2 + \dots + (2k + 1)^2 + (2(k + 1) + 1)^2 = ((k + 1) + 1)(2(k + 1) + 1)\frac{2(k+1)+3}{3}$.

From the inductive hypothesis, we assume that $1^2 + 3^2 + 5^2 + \dots + (2k + 1)^2 = (k + 1)(2k + 1)\frac{2k+3}{3}$. Then,

$$\begin{aligned} 1^2 + 3^2 + 5^2 + \dots + (2k + 1)^2 + (2(k + 1) + 1)^2 &= (k + 1)(2k + 1)\frac{2k+3}{3} + (2(k + 1) + 1)^2 \\ &= (k + 1)(2k + 1)\frac{2k+3}{3} + (2k + 3)^2 \\ &= (k + 1)(2k + 1)\frac{2k+3}{3} + (2k + 3)(2k + 3) \\ &= (2k + 3)\left(\frac{(k+1)(2k+1)}{3} + 2k + 3\right) \\ &= (2k + 3)\left(\frac{2k^2+3k+1+6k+9}{3}\right) \\ &= (2k + 3)\left(\frac{2k^2+9k+10}{3}\right) \\ &= (2k + 3)\left(\frac{(k+2)(2k+5)}{3}\right) \\ &= ((k + 1) + 1)(2(k + 1) + 1)\frac{2(k+1)+3}{3}. \end{aligned}$$

Thus, $1^2 + 3^2 + 5^2 + \dots + (2k + 1)^2 + (2(k + 1) + 1)^2 = ((k + 1) + 1)(2(k + 1) + 1)\frac{2(k+1)+3}{3}$ and by the principle of mathematical induction, $1^2 + 3^2 + 5^2 + \dots + (2n + 1)^2 = (n + 1)(2n + 1)\frac{2n+3}{3}$ for all positive integers n ■.

Question 3

Prove that $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = (n+1)! - 1$ whenever n is a positive integer.

Base Case: $P(1)$ is the statement that $1 \cdot 1! = (1+1)! - 1$.

$1 \cdot 1! = 1 = (1+1)! - 1 = 2! - 1 = 2 - 1 = 1 \implies P(1)$ is true.

Inductive Hypothesis: The inductive hypothesis is that $1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! = (k+1)! - 1$ for some positive integer k .

In the inductive step, we need to prove that $1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! + (k+1)(k+1)! = ((k+1)+1)! - 1$.

From the inductive hypothesis, we assume that $1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! = (k+1)! - 1$. Then,

$$\begin{aligned} 1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! + (k+1)(k+1)! &= (k+1)! - 1 + (k+1)(k+1)! \\ &= (k+1)!(1 + k+1) \\ &= (k+1)!(k+2) \\ &= ((k+1)+1)! - 1. \end{aligned}$$

Thus, $1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! + (k+1)(k+1)! = ((k+1)+1)! - 1$.

Therefore, by the principle of mathematical induction, $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = (n+1)! - 1$ for all positive integers n ■.

Question 4

Prove that $3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^n = \frac{3(5^{n+1}-1)}{4}$ whenever n is a nonnegative integer.

Base Case: $P(0)$ is the statement that $3 = \frac{3(5^{0+1}-1)}{4}$.

$$3^0 = 3 = \frac{3(5^{0+1}-1)}{4} = \frac{3(5-1)}{4} = \frac{3(4)}{4} = 3 \implies P(0) \text{ is true.}$$

Inductive Hypothesis: The inductive hypothesis is that $3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^k = \frac{3(5^{k+1}-1)}{4}$ for some nonnegative integer k .

In the inductive step, we need to prove that $3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^k + 3 \cdot 5^{k+1} = \frac{3(5^{k+1+1}-1)}{4}$.

From the inductive hypothesis, we assume that $3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^k = \frac{3(5^{k+1}-1)}{4}$. Then,

$$\begin{aligned} 3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^k + 3 \cdot 5^{k+1} &= \frac{3(5^{k+1}-1)}{4} + 3 \cdot 5^{k+1} \\ &= \frac{3(5^{k+1}-1)}{4} + 3 \cdot 5^{k+1} \\ &= \frac{3(5^{k+1}-1)}{4} + \frac{12 \cdot 5^{k+1}}{4} \\ &= \frac{3 \cdot 5^{k+1} - 3 + 12 \cdot 5^{k+1}}{4} \\ &= \frac{15 \cdot 5^{k+1} - 3}{4} \\ &= \frac{3(5^{k+2}-1)}{4}. \end{aligned}$$

Thus, $3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^k + 3 \cdot 5^{k+1} = \frac{3(5^{k+2}-1)}{4}$.

Therefore, by the principle of mathematical induction, $3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^n = \frac{3(5^{n+1}-1)}{4}$ for all nonnegative integers n ■.

Question 5

Prove that $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^n = \frac{(1-(-7)^{n+1})}{4}$ whenever n is a nonnegative integer.

Base Case: $P(0)$ is the statement that $2 = \frac{(1-(-7)^{0+1})}{4}$.

$$2 \cdot 7^0 = 2 = \frac{(1-(-7)^{0+1})}{4} = \frac{(1-(-7)^1)}{4} = \frac{(1-(-7))}{4} = \frac{(1+7)}{4} = \frac{8}{4} = 2 \implies P(0) \text{ is true.}$$

Inductive Hypothesis: The inductive hypothesis is that $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^k = \frac{(1-(-7)^{k+1})}{4}$ for some nonnegative integer k .

In the inductive step, we need to prove that $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^k + 2(-7)^{k+1} = \frac{(1-(-7)^{k+1+1})}{4}$.

From the inductive hypothesis, we assume that $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^k = \frac{(1-(-7)^{k+1})}{4}$. Then,

$$\begin{aligned} 2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^k + 2(-7)^{k+1} &= \frac{(1-(-7)^{k+1})}{4} + 2(-7)^{k+1} \\ &= \frac{(1-(-7)^{k+1})}{4} + 2(-7)^{k+1} \\ &= \frac{(1-(-7)^{k+1})}{4} + \frac{8(-7)^{k+1}}{4} \\ &= \frac{1-(-7)^{k+1}+8(-7)^{k+1}}{4} \\ &= \frac{1+7(-7)^{k+1}}{4} \\ &= \frac{1+(-7)^{k+2}}{4}. \end{aligned}$$

Thus, $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^k + 2(-7)^{k+1} = \frac{(1-(-7)^{k+2})}{4}$.

Question 6

a) Find a formula for the sum of the first n even positive integers.

- For $n = 1$, the sum is 2.
- For $n = 2$, the sum is $2 + 4 = 6$.
- For $n = 3$, the sum is $2 + 4 + 6 = 12$.
- For $n = 4$, the sum is $2 + 4 + 6 + 8 = 20$.

Conjecture: The sum of the first n even positive integers is $2 + 4 + 6 + \dots + 2n = 2(1 + 2 + 3 + \dots + n) = 2\frac{n(n+1)}{2} = n(n+1)$.

b) Prove the formula that you conjectured in part (a). Let $P(n)$ be the statement that $2 + 4 + 6 + \dots + 2n = n(n+1)$ for the positive integer n .

Base Case: $P(1)$ is the statement that $2 = 1(1+1) = 2$.

$P(k)$ is the statement that $2 + 4 + 6 + \dots + 2k = k(k+1)$.

Inductive Hypothesis: Assume that $2 + 4 + 6 + \dots + 2k = k(k+1)$.

Then,

$$\begin{aligned} 2 + 4 + 6 + \dots + 2k + 2(k+1) &= k(k+1) + 2(k+1) \\ &= k^2 + k + 2k + 2 \\ &= k^2 + 3k + 2 \\ &= (k+1)(k+2). \end{aligned}$$

Thus, $2 + 4 + 6 + \dots + 2k + 2(k+1) = (k+1)(k+2)$.

Therefore, by the principle of mathematical induction, $2 + 4 + 6 + \dots + 2n = n(n+1)$ for all positive integers n ■.

Question 7

a) Find a formula for $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + (\frac{1}{2})^n$ by examining the values of this expression for small values of n . (Hint: use geometric progression to make a good guess.)

- For $n = 1$, the sum is $1/2 + 1/4 = 3/4$.
- For $n = 2$, the sum is $1/2 + 1/4 + 1/8 = 7/8$.
- For $n = 3$, the sum is $1/2 + 1/4 + 1/8 + 1/16 = 15/16$.

Conjecture: The sum of the first n terms is $1 - (1/2)^n$.

b) Prove the formula you conjectured in part (a). Let $P(n)$ be the statement that $1/2 + 1/4 + 1/8 + \dots + (1/2)^n = 1 - (1/2)^n$ for the positive integer n .

Base Case: $P(1)$ is the statement that $1/2 = 1 - (1/2) = 1/2$.

Inductive Hypothesis: Assume that $1/2 + 1/4 + 1/8 + \dots + (1/2)^k = 1 - (1/2)^k$.

Then,

$$\begin{aligned} 1/2 + 1/4 + 1/8 + \dots + (1/2)^k + (1/2)^{k+1} &= (1 - (1/2)^k) + (1/2)^{k+1} \\ &= 1 - (1/2)^k + (1/2)^{k+1} \\ &= 1 - (1/2)^k + (1/2)(1/2)^k \\ &= 1 - (1/2)^k(1 - 1/2) \\ &= 1 - (1/2)^k(1/2) \\ &= 1 - (1/2)^{k+1}. \end{aligned}$$

Thus, $1/2 + 1/4 + 1/8 + \dots + (1/2)^k + (1/2)^{k+1} = 1 - (1/2)^{k+1}$.

Therefore, by the principle of mathematical induction, $1/2 + 1/4 + 1/8 + \dots + (1/2)^n = 1 - (1/2)^n$ for all positive integers n ■.

Question 8

a) Find a formula for $1/3 + 1/9 + 1/27 + \dots + \frac{1}{3^n}$ by examining the values of this expression for small values of n . (Hint: use geometric progression to make a good guess.)

- For $n = 1$, the sum is $1/3 = 1/3$.
- For $n = 2$, the sum is $1/3 + 1/9 = 4/9$.
- For $n = 3$, the sum is $1/3 + 1/9 + 1/27 = 13/27$.

Conjecture: The sum of the first n terms is $\frac{1}{2}(1 - \frac{1}{3^n})$.

b) Prove the formula you conjectured in part (a). Let $P(n)$ be the statement that $1/3 + 1/9 + 1/27 + \dots + \frac{1}{3^n} = \frac{1}{2}(1 - \frac{1}{3^n})$ for the positive integer n .

Inductive Hypothesis: Assume that $1/3 + 1/9 + 1/27 + \dots + \frac{1}{3^k} = \frac{1}{2}(1 - \frac{1}{3^k})$.

Then,

$$\begin{aligned} 1/3 + 1/9 + 1/27 + \dots + \frac{1}{3^k} + \frac{1}{3^{k+1}} &= \frac{1}{2}(1 - \frac{1}{3^k}) + \frac{1}{3^{k+1}} \\ &= \frac{1}{2}(1 - \frac{1}{3^k}) + \frac{1}{3^{k+1}} \\ &= \frac{1}{2}(1 - \frac{1}{3^k}) + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{3^k} \\ &= \frac{1}{2}(1 - \frac{1}{3^k} + \frac{2}{3} \cdot \frac{1}{3^k}) \\ &= \frac{1}{2}(1 - \frac{1}{3^k} + \frac{2}{3} \cdot \frac{1}{3^k}) \\ &= \frac{1}{2}(1 - \frac{1}{3} \cdot \frac{1}{3^k}) \\ &= \frac{1}{2}(1 - \frac{1}{3^{k+1}}). \end{aligned}$$

Thus, $1/3 + 1/9 + 1/27 + \dots + \frac{1}{3^k} + \frac{1}{3^{k+1}} = \frac{1}{2}(1 - \frac{1}{3^{k+1}})$.

Therefore, by the principle of mathematical induction, $1/3 + 1/9 + 1/27 + \dots + \frac{1}{3^n} = \frac{1}{2}(1 - \frac{1}{3^n})$ for all positive integers n ■.

Question 9

Prove that $1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1}n^2 = (-1)^{n-1}n\frac{n+1}{2}$ whenever n is a positive integer.

Base Case: $P(1)$ is the statement that $1^2 = (-1)^{1-1}1\frac{1+1}{2}$.

$$1^2 = 1 = (-1)^{1-1}1\frac{1+1}{2} = 1 \cdot 1 \cdot 1 = 1 \implies P(1) \text{ is true.}$$

Inductive Hypothesis: The inductive hypothesis is that $1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1}k^2 = (-1)^{k-1}k\frac{k+1}{2}$ for some positive integer k .

In the inductive step, we need to prove that $1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1}k^2 + (-1)^k(k+1)^2 = (-1)^k(k+1)\frac{k+1+1}{2}$.

By the inductive hypothesis, we assume that $1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1}k^2 = (-1)^{k-1}k\frac{k+1}{2}$. Then,

$$\begin{aligned} 1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1}k^2 + (-1)^k(k+1)^2 &= (-1)^{k-1}k\frac{k+1}{2} + (-1)^k(k+1)^2 \\ &= (-1)^{k-1}k\frac{k+1}{2} + (-1)^k(k+1)^2 \\ &= (-1)^{k-1}k\frac{k+1}{2} + (-1)(-1)^{k-1}(k+1)^2 \\ &= (-1)^{k-1}k\frac{k+1}{2} - (k+1)^2 \\ &= (-1)^{k-1}k\frac{k+1}{2} - (k+1)(k+1) \\ &= (-1)^{k-1}(k+1)\left(\frac{k}{2} - (k+1)\right) \\ &= (-1)^{k-1}(k+1)\left(\frac{k-2k-2}{2}\right) \\ &= (-1)^{k-1}(k+1)\left(\frac{-k-2}{2}\right) \\ &= (-1)^k(k+1)\left(\frac{-k-2}{2}\right) \\ &= (-1)^k(k+1)\frac{k+1+1}{2}. \end{aligned}$$

Thus, $1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1}k^2 + (-1)^k(k+1)^2 = (-1)^k(k+1)\frac{k+1+1}{2}$.

Therefore, by the principle of mathematical induction, $1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1}n^2 = (-1)^{n-1}n\frac{n+1}{2}$ for all positive integers n ■.

Question 10

Let $P(n)$ be the statement that a postage of n cents can be formed using just 3-cent stamps and 5-cent stamps. The parts of this exercise outline a strong induction proof that $P(n)$ is true for $n \geq 8$.

a) Show that the statements $P(8)$, $P(9)$, and $P(10)$ are true, completing the basis step of the proof.

- $P(8)$ is the statement that a postage of 8 cents can be formed using just 3-cent stamps and 5-cent stamps. This is true because $8 = 3 + 5$.
- $P(9)$ is the statement that a postage of 9 cents can be formed using just 3-cent stamps and 5-cent stamps. This is true because $9 = 3 + 3 + 3$.
- $P(10)$ is the statement that a postage of 10 cents can be formed using just 3-cent stamps and 5-cent stamps. This is true because $10 = 5 + 5$.

b) What is the inductive hypothesis of the proof? What do you need to prove in the inductive step? Inductive Hypothesis: The inductive hypothesis is that $P(k)$ is true for some positive integer k . In the inductive step, we need to prove that $P(k + 1)$ is true.

c) Complete the inductive step for $k \geq 10$. Explain why these steps show that this statement is true whenever $n \geq 8$. By the inductive hypothesis, we assume that $P(k)$ is true for some positive integer $k \geq 10$. Then, $k = 3a + 5b$ for some nonnegative integers a and b . Then, $k + 1 = 3(a - 2) + 5(b + 2)$, which is true because $a - 2$ and $b + 2$ are nonnegative integers. Thus, $P(k + 1)$ is true. Therefore, by the principle of mathematical induction, $P(n)$ is true for all positive integers $n \geq 8$.

Question 11

Let $P(n)$ be the statement that a postage of n cents can be formed using just 4-cent stamps and 7-cent stamps. The parts of this exercise outline a strong induction proof that $P(n)$ is true for $n \geq 18$.

a) Show statements $P(18)$, $P(19)$, $P(20)$, and $P(21)$ are true, completing the basis step of the proof.

- $P(18)$ is the statement that a postage of 18 cents can be formed using just 4-cent stamps and 7-cent stamps. This is true because $18 = 4 + 7 + 7$.
- $P(19)$ is the statement that a postage of 19 cents can be formed using just 4-cent stamps and 7-cent stamps. This is true because $19 = 4 + 4 + 4 + 7$.
- $P(20)$ is the statement that a postage of 20 cents can be formed using just 4-cent stamps and 7-cent stamps. This is true because $20 = 4 + 4 + 4 + 4 + 4$.

b) What is the inductive hypothesis of the proof? What do you need to prove in the inductive step? Inductive Hypothesis: The inductive hypothesis is that $P(k)$ is true for some positive integer $k \geq 21$. In the inductive step, we need to prove that $P(k+1)$ is true.

c) Complete the inductive step for $k \geq 21$. Explain why these steps show that this statement is true whenever $n \geq 18$. By the inductive hypothesis, we assume that $P(k)$ is true for some positive integer $k \geq 21$.

Inductive Step: Assume $P(k)$ is true for all $18 \leq k \leq n$, where $n \geq 21$. We want to show that $P(n+1)$ is true.

Since $n \geq 21$, we know that $n-3 \geq 18$. Therefore, by the inductive hypothesis, $P(n-3)$ is true, meaning we can form a postage of $n-3$ cents using 4-cent and 7-cent stamps.

To form a postage of $n+1$ cents, simply add one more 4-cent stamp to the postage of $n-3$ cents. Thus, $P(n+1)$ is true. Therefore, by the principle of mathematical induction, $P(n)$ is true for all positive integers $n \geq 18$.

This shows that the statement is true whenever $n \geq 18$ because the basis step shows that the statement is true for $n = 18, 19, 20, 21$ and the inductive step shows that if the statement is true for $n = k$ where $k \geq 21$, then the statement is true for $n = k+1$.

Question 12

a) Determine which amounts of postage can be formed using just 4-cent and 11-cent stamps. The amounts of postage that can be formed using just 4-cent and 11-cent stamps are the multiples of the greatest common divisor of 4 and 11, which is 1. Thus, all amounts of postage can be formed using just 4-cent and 11-cent stamps.

b) Prove your answer to (a) using the principle of mathematical induction. Be sure to state explicitly your inductive hypothesis in the inductive step. Let $P(n)$ be the statement that a postage of n cents can be formed using just 4-cent and 11-cent stamps. We will prove that $P(n)$ is true for all positive integers n .

Base Case: $P(1)$ is the statement that a postage of 1 cent can be formed using just 4-cent and 11-cent stamps. This is true because $1 = 11 - 4$.

Inductive Hypothesis: Assume that $P(k)$ is true for some positive integer k . That is, $k = 4a + 11b$ for some nonnegative integers a and b .

$P(k+1)$ is the statement that a postage of $k+1$ cents can be formed using just 4-cent and 11-cent stamps. Then, $k+1 = 4(a-11) + 11(b+4)$, which is true because $a-11$ and $b+4$ are nonnegative integers. Thus, $P(k+1)$ is true.

Therefore, by the principle of mathematical induction, $P(n)$ is true for all positive integers n ■.

c) Prove your answer to (a) using strong induction. How does the inductive hypothesis in this proof differ from that in the inductive hypothesis for a proof using mathematical induction? Let $P(n)$ be the statement that a postage of n cents can be formed using just 4-cent and 11-cent stamps. We will prove that $P(n)$ is true for all positive integers n .

Base Case: $P(1)$ is the statement that a postage of 1 cent can be formed using just 4-cent and 11-cent stamps. This is true because $1 = 11 - 4$.

Inductive Hypothesis: Assume that $P(1), P(2), \dots, P(k)$ are true for some positive integer k . That is, $1, 2, \dots, k$ cents can be formed using just 4-cent and 11-cent stamps. Then, $k+1 = 4(a-11) + 11(b+4)$, which is true because $a-11$ and $b+4$ are nonnegative integers. Thus, $P(k+1)$ is true.

Therefore, by the principle of strong induction, $P(n)$ is true for all positive integers n ■.

The inductive hypothesis in this proof differs from that in the inductive hypothesis for a proof using mathematical induction in that the inductive hypothesis in this proof assumes that $P(1), P(2), \dots, P(k)$ are true for some positive integer k , whereas the inductive hypothesis in the inductive hypothesis for a proof using mathematical induction assumes that $P(k)$ is true for some positive integer k .

Question 13

a) Determine which amounts of postage can be formed using just 3-cent and 10-cent stamps. The amounts of postage that can be formed using just 3-cent and 10-cent stamps are the multiples of the greatest common divisor of 3 and 10, which is 1. Thus, all amounts of postage can be formed using just 3-cent and 10-cent stamps.

b) Prove your answer to (a) using the principle of mathematical induction. Be sure to state explicitly your inductive hypothesis in the inductive step. Let $P(n)$ be the statement that a postage of n cents can be formed using just 3-cent and 10-cent stamps. We will prove that $P(n)$ is true for all positive integers n .

Base Case: $P(1)$ is the statement that a postage of 1 cent can be formed using just 3-cent and 10-cent stamps. This is true because $1 = 10 - 3 - 3 - 3$.

Inductive Hypothesis: Assume that $P(k)$ is true for some positive integer k . That is, $k = 3a + 10b$ for some nonnegative integers a and b .

$P(k+1)$ is the statement that a postage of $k+1$ cents can be formed using just 3-cent and 10-cent stamps. Then, $k+1 = 3(a-10) + 10(b+3)$, which is true because $a-10$ and $b+3$ are nonnegative integers. Thus, $P(k+1)$ is true.

Therefore, by the principle of mathematical induction, $P(n)$ is true for all positive integers n ■.

c) Prove your answer to (a) using strong induction. How does the inductive hypothesis in this proof differ from that in the inductive hypothesis for a proof using mathematical induction? Let $P(n)$ be the statement that a postage of n cents can be formed using just 3-cent and 10-cent stamps. We will prove that $P(n)$ is true for all positive integers n .

Base Case: $P(1)$ is the statement that a postage of 1 cent can be formed using just 3-cent and 10-cent stamps. This is true because $1 = 10 - 3 - 3 - 3$.

Inductive Hypothesis: Assume that $P(1), P(2), \dots, P(k)$ are true for some positive integer k . That is, $1, 2, \dots, k$ cents can be formed using just 3-cent and 10-cent stamps. Then, $k+1 = 3(a-10) + 10(b+3)$, which is true because $a-10$ and $b+3$ are nonnegative integers. Thus, $P(k+1)$ is true.

Therefore, by the principle of strong induction, $P(n)$ is true for all positive integers n ■.

The inductive hypothesis in this proof differs from that in the inductive hypothesis for a proof using mathematical induction in that the inductive hypothesis in this proof assumes that $P(1), P(2), \dots, P(k)$ are true for some positive integer k , whereas the inductive hypothesis in the inductive hypothesis for a proof using mathematical induction assumes that $P(k)$ is true for some positive integer k .

Question 14

Which amounts of money can be formed using just two dollar bills and five-dollar bills? Prove your answer using strong induction.

Theorem: Any amount of money $n \geq 4$ can be formed using two-dollar and five-dollar bills.

Base Cases:

- $P(4)$: $4 = 2 + 2$
- $P(5)$: $5 = 5$

Inductive Hypothesis: Assume that $P(k)$ is true for some positive integer k where $k \geq 5$. That is, $k = 2a + 5b$ for some nonnegative integers a and b .

Inductive Step: We need to prove that $P(k + 1)$ is true.

- Case 1: If k is even, then $k + 1 = 2(a + 1) + 5b$. Since $a + 1$ is a nonnegative integer, $P(k + 1)$ is true.
- Case 2: If k is odd, then $k + 1 = 2(a - 2) + 5(b + 2)$. Since $a - 2$ and $b + 2$ are nonnegative integers, $P(k + 1)$ is true.

In both cases, $P(k + 1)$ is true. Therefore, by the principle of strong induction, $P(n)$ is true for all positive integers $n \geq 4$ ■.

Question 15

Prove that the first player has a winning strategy for the game of Chomp, introduced in a lecture at the class, if the initial board is square. [Hint: Use strong induction to show that this strategy works. For the first move, the first player chomps all cookies except those in the left and top edges. On subsequent moves, after the second player has chomped cookies on either the top or left edge, the first player chomps cookies in the same relative positions on the left or top edge, respectively.]

Let $P(n)$ be the statement that the first player has a winning strategy for the game of Chomp if the initial board is an $n \times n$ square. We will prove that $P(n)$ is true for all positive integers n .

Base Case: $P(1)$ is the statement that the first player has a winning strategy for the game of Chomp if the initial board is a 1×1 square. This is true because the first player can chomp the only cookie on the board.

Inductive Hypothesis: Assume that $P(1), P(2), \dots, P(k)$ are true for some positive integer k . That is, the first player has a winning strategy for the game of Chomp if the initial board is an $n \times n$ square.

Then, the first player chomps all cookies except those in the left and top edges. On subsequent moves, after the second player has chomped cookies on either the top or left edge, the first player chomps cookies in the same relative positions on the left or top edge, respectively. Thus, $P(k + 1)$ is true.

Therefore, by the principle of strong induction, $P(n)$ is true for all positive integers n ■.